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A Study on Factors Influencing Designers' Behavioral Intention in Using AI-Generated Content for Assisted Design: Perceived Anxiety, Perceived Risk, and UTAUT

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ABSTRACT

This study aims to comprehensively understand the intention to use Artificial Intelligence Generated Assistance in Design Tools (AIGC) among design students and practitioners, along with its influencing factors. Utilizing Smart-PLS software and Partial Least Squares Structural Equation Modeling (PLS-SEM) technique, we constructed a comprehensive research model. Based on 404 valid questionnaire responses, we systematically analyzed the underlying mechanisms of designers' attitudes towards AIGC tools. The sample encompasses diverse schools and levels of professional experience, ensuring the wide applicability of research outcomes. In the data analysis process, professional statistical analysis methods, including path analysis and standardized path coefficients, were employed to ensure a profound exploration of research questions. The results indicate that performance expectancy, effort expectancy, social influence, and facilitating conditions significantly positively influence the willingness to use AIGC tools, while perceived anxiety and perceived risk exert negative impacts. This study, by integrating traditional and novel factors, provides crucial theoretical and practical guidance for the actual application of AIGC technology in the design field, offering profound insights for the future development and education of design technology.

KEYWORDS

Artificial intelligence generated content; artificial intelligence; design industry; behavioral intention; UTAUT

1. Introduction

Generative Artificial Intelligence (AI), a subset focusing on creating novel content, demonstrates tremendous potential in various applications, fundamentally transforming human-technology interactions and problem-solving approaches (Russell, 2016). Recent surveys highlight significant advancements in AI-generated content (Xu et al., 2023). The intersection of the Internet of Things and AI sparks unprecedented interest in AI-generated content (AIGC) technology, showcasing incredible capabilities for automated content generation, such as writing scripts and creating social media posts (Xu et al., 2023). This technology permeates various industries, underscoring the importance of understanding and harnessing its potential and challenges for driving future innovations (Mnih et al., 2015).

In the field of art and design, AIGC has gained widespread application (Cao et al., 2023), with scholars discussing its potential applications in artistic technology (Cetinic & She, 2022; Hong & Curran, 2019). Propelled by technological developments, AIGC applications in the field of painting continue to mature (Liu, 2020) Since Google's development of AI for graphic design in 2016 (Wu, 2020), AIGC-assisted design tools have become increasingly prevalent in the creative and design industry. These tools use text-to-image datasets to generate images from textual descriptions, exhibiting various functions such as creating anthropomorphic versions of animals and objects, combining unrelated concepts in seemingly rational ways, rendering text, and applying transformations to existing images (Thanigan et al., 2021). This study attempts a practical verification through the generation of images using the Midjourney software, as depicted in Figure 1. The efficiency, precision, and creativity of these intelligent tools attract an increasing number of designers. Behind AIGC-assisted design tools, the integration of the Technology Acceptance Model provides a crucial theoretical framework, assisting us in understanding and explaining user attitudes and behavioral intentions towards the adoption of new technology (Gmeiner et al., 2023). The efficiency, precision, and creativity of these intelligent tools attract an increasing number of designers. Behind AIGC-assisted design tools, the integration of the Technology Acceptance Model provides a crucial theoretical framework, assisting us in understanding and explaining user attitudes and behavioral intentions towards the adoption of new technology.

While factors such as performance expectancy, effort expectancy, facilitating conditions, and social influence have been extensively studied and thoroughly validated within the Technology Acceptance Model (Kandoth & Shekhar, 2022; Dave et al., 2023), new factors may influence the behavioral intentions of designers when confronted with emerging technologies like AIGC. Therefore, this study, building upon the Technology Acceptance Model, introduces the dimensions of perceived anxiety and perceived risk. Perceived anxiety refers to an individual's concerns and apprehensions about potential negative impacts resulting from the introduction of new

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Figure 1. Images generated by AIGC tool: generated by midjournal software in this study.

technology or methods (Reddy et al., 2021), while perceived risk is the expectation of potential problems following the introduction of technology or methods (Lee, 2009). The aim is to comprehensively explore designers' attitudes and adoption mechanisms towards AIGC-assisted design tools. The research objective is to gain in-depth insights into designers' behavioral intentions regarding AIGC-assisted design tools, facilitating continuous development in this field. Through structural equation model analysis, incorporating performance expectancy, effort expectancy, social influence, facilitating conditions, perceived anxiety, and perceived risk into the research framework, this study aims to provide new insights and revelations for academic research and practical application in related domains.

In the following sections, we will first review past literature related to AIGC-assisted design tools and designers' behavioral intentions. Subsequently, we will detail the methodology and data analysis of our study, followed by reporting and discussing research findings. Finally, we will summarize our research findings and propose prospects for future research. Through these efforts, we aim to deepen the understanding of designers' attitudes and behavioral intentions towards AIGC-assisted design tools, providing theoretical foundations and practical guidance for the widespread application of AIGC technology in the design field. Simultaneously, by incorporating perceived anxiety and perceived risk into our study, we aim to dissect designers' concerns and apprehensions about new technology, offering recommendations for the improvement and optimization of AIGC-assisted design tools to relevant enterprises and institutions.

2. Theoretical support and research hypotheses

2.1. Variables under the unified theory of acceptance and use of technology (UTAUT): Performance expectancy, effort expectancy, social influence, facilitating conditions, and behavioral intention

The Unified Theory of Acceptance and Use of Technology (UTAUT) is a widely applied model for studying user adoption behavior of information technology. In previous studies related to AI and user usage, UTAUT has been empirically tested for its validity and reliability (Venkatesh, 2022; Zha, 2020; Lin et al., 2022). UTAUT serves as a significant theoretical framework in the context of designers' behavioral intentions towards AI-generated content (AIGC) assisted design tools. Firstly, the constructs of performance expectancy, effort expectancy, social influence, and facilitating conditions have been well-validated in UTAUT (Venkatesh et al., 2003), demonstrating their significant impact on user technology adoption behavior in various fields (Chang, 2012; Chao, 2019). In the context of designers' acceptance of AIGC-assisted design tools, these constructs possess strong explanatory power for predicting designers' adoption and acceptance of this technology. Secondly, UTAUT provides a concise and effective theoretical framework to understand users' attitudes and behavioral intentions towards technology (Lu & Yang, 2014).

In this study, the UTAUT model plays a crucial role in explaining and analyzing designers' willingness to use AIGC-assisted design tools. By integrating key constructs from these models, the study can gain a more comprehensive understanding of designers' attitudes and intentions towards new technology. Based on this, the following hypotheses are proposed:

H1: Performance expectancy positively influences designers' behavioral intention.

H2: Effort expectancy positively influences designers' behavioral intention.

H3: Facilitating conditions positively influence designers' behavioral intention.

H4: Social influence positively influences designers' behavioral intention.

Social influence refers to the impact of others on an individual's adoption of a specific technology (Venkatesh et al., 2003). In the design domain, designers often enhance their skills and knowledge through communication and discussion with peers, colleagues, and other professionals. Literature

suggests that social influence plays a crucial role in information acquisition and knowledge sharing during the technology adoption process (Slade et al., 2015). Some studies found a positive correlation between social influence and facilitating conditions (Pratama, 2021). Through interaction and exchange with other designers in the community, designers may gain more insights into the usage techniques and advantages of AIGC-assisted design tools, thus increasing their awareness of the convenience of using these tools. Moreover, social influence can also promote technology adoption through positive social support and encouragement (Wu & Chen, 2015). When designers observe successful applications of AIGC technology among their peers in the community, they may be more motivated to try using this tool, believing they can also benefit from it, thereby increasing their perception of the convenience of AIGC-assisted design tools. Research indicates that social influence positively influences performance expectancy (Al Qeisi & Al-Abdallah, 2014; Maillet et al., 2015; Jiao et al., 2021) as well. In the social network of the design field, the sharing of successful cases and excellent design works can inspire other designers to strive for higher performance. Through the sharing of experiences and interactions within the community, designers can understand the potential advantages of AIGC-assisted design tools, thereby forming expectations for their improvement in design performance. Based on this, the following hypotheses are proposed:

H5: Social influence positively influences facilitating conditions.

H6: Social influence positively influences performance expectancy.

Facilitating conditions refer to consumers' perceived availability of resources and support for executing technical behavior (Venkatesh et al., 2003; Bendary & Al-Sahouly, 2018) In the adoption process of AIGC-assisted design tools by designers, the perceived ease of use significantly impacts their usage. Studies suggest that the usability and learning curve of technology are crucial factors influencing users' willingness to invest effort in learning and using it (Kumar et al., 2020; Rezvani et al., 2022) If the learning curve of AIGC-assisted design tools is relatively gentle, designers are more likely to pick up and master the technology, increasing their effort expectancy. Similarly, effort expectancy positively influences performance expectancy (Abou-Shouk & Soliman, 2021; Al Qeisi & Al-Abdallah, 2014). Specifically, when designers use AIGC-assisted design tools, they anticipate whether putting in more effort will result in better performance. Based on the theoretical framework of the Unified Theory of Acceptance and Use of Technology, it is hypothesized that the more designers tend to believe that putting in effort will lead to better performance, the more likely they are to expect higher performance when using AIGC technology. Based on this, the following hypotheses are proposed:

H7: Facilitating conditions positively influence effort expectancy.

H8: Effort expectancy positively influences performance expectancy.

2.2. Perceived anxiety and perceived risk

In addition to integrating variables from the Unified Theory of Acceptance and Use of Technology (UTAUT), two new constructs are added: perceived anxiety and perceived risk. This is because, in the process of designers' acceptance of AIGC-assisted design tools, they may face concerns about the future and perceive potential risks associated with the technology. There is literature suggesting that AIGC poses a serious threat to the traditional graphic design industry and designers (Wu, 2020).

Anxiety is defined as an emotional state of fear, discomfort, discouragement, reflection, and apprehension that influences decision-making (Nayak, 2014). Anxiety is related to the fear individuals experience when interacting with underlying technologies, such as the discomfort, perception, and stress events that generate pressure (Patil et al., 2020). Perceived anxiety refers to an individual's concerns and apprehensions about potential negative impacts resulting from the introduction of new technology or methods (Thanigan et al., 2021). In the design field, designers may experience perceived anxiety about the introduction of new technology, as they are concerned that these technologies may affect their creativity, design processes, and career prospects. Perceived anxiety may stem from concerns about future uncertainty (Grupe & Nitschke, 2013), especially in an era of rapid technological development where designers may worry about the ongoing relevance of their skills. Perceived anxiety, through emotional reactions such as uneasiness and stress, can affect designers' decision-making and behavior, making them hesitant to adopt new technology. The perceived anxiety construct is crucial for understanding designers' attitudes towards AIGC-assisted design tools. This anxiety may lead designers to be reluctant to adopt the technology, or even resist its use. Therefore, considering that perceived anxiety is expected to negatively influence behavioral intention (Wu & Wang, 2005; de Sena Abrahão et al., 2016), it is included in the research framework to comprehensively explore designers' attitudes and acceptance.

Perceived risk refers to an individual's expectation of potential problems following the introduction of technology or methods (Lee, 2009) In the design field, designers may perceive risks associated with the use of new technology (Burns et al., 2012). Perceived risk may be associated with negative emotions, such as fear and concern, affecting designers' attitudes and intentions. Designers may believe that if technology poses a risk, it may not be worth the risk to try it. Research indicates that perceived risk negatively influences behavioral intention (Lee & Song, 2013). The perceived risk construct focuses on designers' perception of potential issues related to AIGC-assisted design tools, such as ethical (Wu et al., 2023), copyright, and privacy concerns. Although AIGC technology has potential advantages in improving design efficiency (Wu et al., 2023), the issues it raises in terms of intellectual property protection (Chen, Fu et al., 2023), privacy (Chen, Wu et al., 2023), and ethics cannot be ignored. Designers' perception of these potential risks

may influence their level of acceptance of AIGC technology. Therefore, the perceived risk construct is included to further study designers' concerns and worries about technical risks.

H9: Perceived anxiety negatively influences behavioral intention.

H10: Perceived risk negatively influences behavioral intention.

In this study, the potential impact of perceived anxiety on designers' performance expectancy is explored. Performance expectancy refers to designers' expectations of the extent to which the use of AIGC-assisted design tools can improve their job performance. Research suggests that an increase in perceived anxiety leads to a decrease in performance expectancy (Eysenck & Calvo, 1992), as perceived anxiety involves designers' concerns about unknown variables and potential challenges brought about by new technology, such as fear of technology getting out of control or fear of unemployment. When designers consider adopting AIGC tools, the level of perceived anxiety may negatively impact their expectations for the tools' effectiveness in improving work efficiency and design quality. Specifically, a high level of anxiety may lead designers to be skeptical about whether these tools can effectively improve work efficiency and design quality. Therefore, this study expects a reverse relationship between perceived anxiety and performance expectancy, meaning that an increase in perceived anxiety will decrease performance expectancy. Furthermore, perceived anxiety negatively affects effort expectancy (Celik, 2016). When designers have concerns and anxiety about AIGC-assisted design tools, they may believe that learning and mastering this technology require more effort and feel uncertain about the difficulty of learning and using it. This uncertainty may reduce designers' enthusiasm for learning and mastering AIGC technology, thus reducing their effort expectancy.

H11: Perceived anxiety negatively influences performance expectancy.

H12: Perceived anxiety negatively influences effort expectancy.

Research suggests that social influence positively influences perceived risk (Lovreglio et al., 2016). Social influence may impact designers' perceived risk of AIGC-assisted design tools. Through communication with peers and other professionals, designers may learn about the advantages, disadvantages, and potential risks of AIGC-assisted design tools in practical applications. If other designers in the community have relatively positive evaluations and opinions of AIGC-assisted design tools, designers may be more inclined to accept and adopt this technology, thereby reducing perceived risk. Additionally, research has already demonstrated that perceived risk negatively influences effort expectancy (Chan et al., 2022). The higher the perceived risk of AIGC-assisted design tools by designers, the more they may believe that learning and mastering this technology require greater effort. This perceived risk may lead designers to be less proactive in learning and using AIGC technology, thereby affecting their effort expectancy.

H13: Social influence influences perceived risk.

H14: Perceived risk influences effort expectancy.

Based on the above hypotheses, the research hypothesis diagram is as follows (Figure 2).

3. Methodology

3.1. Research design

This study aims to gain a comprehensive understanding of the behavioral intentions of design students and professionals towards AI-generated content (AIGC) assisted design tools and the factors influencing these intentions. To achieve this, we have carefully designed a data collection process to build a comprehensive database for analyzing and understanding the attitudes and viewpoints of the target group. The study includes design students from various schools and professionals with practical design experience. The sample design aims to reflect diversity in educational backgrounds and levels of professional experience to ensure the broad applicability and depth of the results.

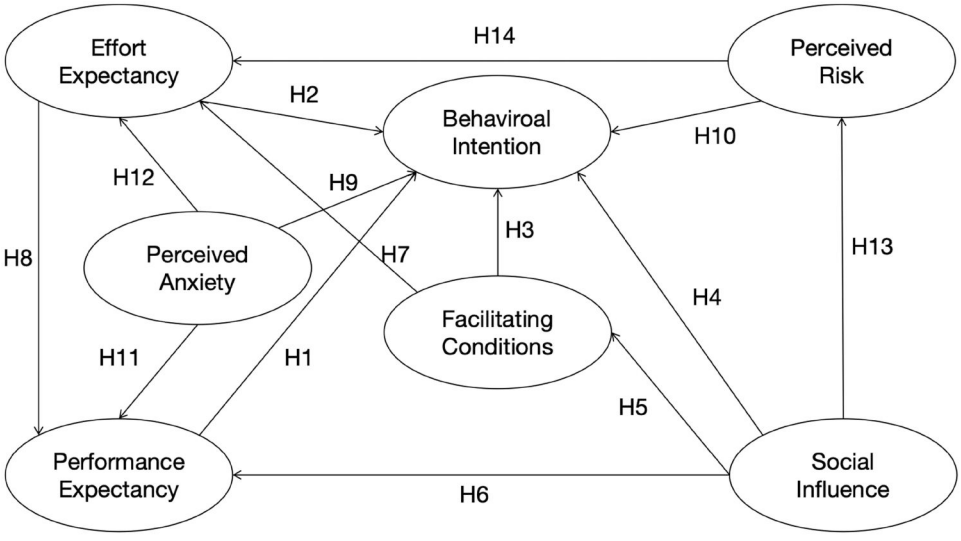


Figure 2. Hypothetical model.

3.2. Analysis techniques

In the data analysis process, this study employs Smart-PLS software to construct and evaluate the structural equation model using Partial Least Squares Structural Equation Modeling (PLS-SEM) technique to handle the data. This method allows effective handling of complex relationships between latent variables, enabling accurate evaluation of model paths and the strength of associations between variables.

3.3. Data collection

For design students and professionals, this study designed a questionnaire survey comprising multiple constructs. The questionnaire includes measurement items for constructs such as performance expectancy, effort expectancy, social influence, facilitating conditions, perceived anxiety, perceived risk, and behavioral intention. Measurement items utilize existing scales, adjusted and validated appropriately to ensure their applicability and reliability. The behavioral measurement scale is detailed in Table 1. To achieve the research objectives, the research team randomly selected participants from design students and professionals in different regions. The diversity of the sample aids in obtaining comprehensive and representative data. The questionnaire was distributed through online survey platforms in July 2023 to ensure efficient and accurate data collection. Participants answered questions based on their experience and attitudes towards AIGC-assisted design tools. A total of 483 responses were collected, and after removing invalid questionnaires, the effective sample size was 404, with an effective response rate of 83.6%.

The survey questionnaire specifically targets individuals in the design profession and industry. The respondents cover a diverse range of individuals at various levels. The percentage distribution of respondents is shown in Table 2.

4. Data analysis

4.1. Normality testing

Prior to conducting the measurement and structural model analyses, a preliminary normality test was performed on each variable item. Data were considered to exhibit normality when the kurtosis and skewness values fell within the range of |10| and |3|, respectively (Moorthy et al., 2019). The kurtosis and skewness values for the data ranged from −1.724 to 1.823 and −1.491 to 0.432, as presented in Table 3. Based on these results, all available data demonstrated a relatively normal distribution.

4.2. 测试模型试验

The validity test of the measurement model was observed in the content, convergent, and discriminant validities (Cheng & Tsai, 2020; Habibi et al., 2020). Measurement items were drawn from existing literature and subsequently subjected to a preliminary survey. The results indicated satisfactory content validity of the model. Additionally, Table 4 demonstrates that

Table 1. Behavioral measurement scale.

No	Construct	Items	Original reference
1	Performance expectancy	PE1 Using AI drawing tools improves my work efficiency. PE2 Using AI drawing tools can enhance the convenience of my work. PE3 Using AI drawing tools is helpful for generating creative inspiration.	(Birch & Irvine, 2009; Wrycza et al., 2017; Chiu et al., 2022)
2	Effort expectancy	EE1 Learning to use AI drawing tools in the design process is easy for me. EE2 The human-computer interaction in using AI drawing tools is clear and understandable. EE3 Proficiently using AI drawing tools in the design process is easy for me.	(Zou et al., 2022)
3	Facilitating conditions	FC1 I have support for the hardware devices used in AI drawing tools. FC2 I have support for the network conditions used in AI drawing tools. FC3 If I need help, I can find someone to assist me in using AI drawing tools.	(Anouze & Alamro, 2019; Zou et al. 2022)
4	Social influence	SI1 People who influence my behavior think I should use AI drawing tools in the professional field. SI2 Important people to me think I should use AI drawing tools in the professional field. SI3 My teachers strongly support me using AI drawing tools in the professional field.	(Chuang, 2016; Zou et al., 2022)
5	Perceived anxiety	PA1 I am worried that the design industry will increasingly rely on AI drawing technology in the future. PA2 I am worried that not learning AI drawing technology will threaten my job position (or future job search). PA3 I am worried that AI drawing will bring about technological changes and changes in the job market.	(Paredes et al., 2021)
6	Perceived risk	PR1 I am worried that the content or data provided before AI drawing is leaked and disclosed. PR2 I am worried that the content generated by AI drawing may involve copyright issues that I am not aware of. PR3 I am worried that using AI drawing tools may have ethical issues.	(Bekaert & Harvey, 2003)
7	Behavioral intention	BI1 I plan to continue using AI drawing tools rather than stopping. BI2 In the future, I will frequently use AI drawing tools in work or professional learning. BI3 I will recommend using AI drawing tools to friends.	(Chen & Chen, 2009; Zou et al., 2022)

Table 2. Demographic characteristics of the respondents.

Attribute	Option	Frequency	Percentage (%)
Gender	Male	185	45.79
	Female	219	54.21
Age (years)	18–25	350	86.63
	26–40	54	13.37
Major background	Visual communication design	320	79.21
	Industrial design	13	3.22
	Environmental design	16	3.96
	Landscape design	8	1.98
	Animation design	10	2.48
	Fashion design	11	2.72
	Product design	17	4.21
	Stage design	9	2.23
Occupation	Undergraduate or college student	326	80.69
	Graduate student (Master's or Ph.D.)	50	12.38
	Design professional	28	6.93
Experience	Yes	360	89.11
	No	44	10.89

Table 3. Descriptive statistics and normality testing.

Construct	Code	Minimum	Maximum	Mean	Standard deviation	Median	Kurtosis	Skewness
PE	PE1	2.000	7.000	5.772	1.522	6.000	0.893	−1.416
	PE2	2.000	7.000	5.849	1.522	6.000	1.108	−1.491
	PE3	2.000	7.000	5.663	1.552	6.000	0.639	−1.345
EE	EE1	2.000	7.000	5.802	1.501	6.000	0.915	−1.402
	EE2	1.000	7.000	5.745	1.605	6.000	0.609	−1.351
	EE3	1.000	7.000	5.770	1.625	6.000	0.816	−1.428
FC	FC1	1.000	7.000	5.304	1.111	5.000	0.998	−0.547
	FC2	1.000	7.000	5.277	1.081	5.000	1.329	−0.664
	FC3	1.000	7.000	5.366	1.125	5.000	1.823	−0.852
SI	SI1	2.000	7.000	5.300	1.893	6.000	−0.981	−0.783
	SI2	1.000	7.000	5.196	1.779	6.000	−0.934	−0.763
	SI3	1.000	7.000	5.213	1.939	6.000	−1.030	−0.766
PA	PA1	1.000	7.000	4.500	2.097	5.000	−1.710	−0.113
	PA2	1.000	7.000	4.510	2.053	5.000	−1.724	−0.069
	PA3	2.000	7.000	4.532	2.010	5.000	−1.694	−0.119
PR	PR1	2.000	7.000	4.161	2.037	3.000	−1.631	0.304
	PR2	1.000	7.000	4.079	2.121	3.000	−1.702	0.284
	PR3	1.000	7.000	3.896	2.121	3.000	−1.591	0.432
BI	BI1	2.000	7.000	5.666	1.622	6.000	0.394	−1.315
	BI2	1.000	7.000	5.634	1.765	6.000	−0.045	−1.198
	BI3	1.000	7.000	5.473	1.930	6.000	−0.560	−1.043

Table 4. Factor loadings, VIF, reliability and validity statistics.

Dimension	Code	VIF	BI	EE	FC	PA	PE	PR	SI	Cronbach A	CR	AVE
Performance expectancy	BI1	4.242	0.945							0.936	0.937	0.886
	BI2	4.553	0.949									
	BI3	3.555	0.929									
Effort expectancy	EE1	4.297		0.942						0.949	0.950	0.907
	EE2	5.080		0.948								
	EE3	6.866		0.968								
Facilitating conditions	FC1	2.948			0.923					0.921	0.926	0.863
	FC2	3.952			0.934							
	FC3	3.589			0.929							
Social influence	PA1	7.979				0.970				0.968	0.968	0.940
	PA2	6.439				0.964						
	PA3	8.958				0.974						
Perceived anxiety	PE1	3.778					0.937			0.927	0.927	0.873
	PE2	3.619					0.934					
	PE3	3.532					0.932					
Perceived risk	PR1	8.903						0.974		0.973	0.973	0.948
	PR2	8.132						0.972				
	PR3	9.042						0.975				
Behaviroal intention	SI1	7.019							0.966	0.958	0.958	0.922
	SI2	4.714							0.950			
	SI3	6.956							0.965			

the factor loading of each measurement variable with its latent variable is more significant than the correlation coefficients with other determining factors (cross-factor loading). This attests to the good convergence and discriminant validity of the measurement model analysis.

Table 4 also presents the values of Cronbach's α, structural reliability, and average variance extracted (AVE). Convergent validity is described as a prerequisite related to the structure of variables. An AVE greater than 0.5 is considered ideal. All observed measurement models exhibited good convergent validity, with the lowest AVE value found for facilitating conditions at 0.863.

Moreover, Internal Consistency Reliability (ICR) was employed to assess the consistency of results across all indicators, where the values of CR (Composite Reliability) and CA (Cronbach's Alpha) should fall between 0 and 1 (Hair et al., 2021). In this context, the model reliability was considered good when CR and CA were not less than 0.7.

Table 5 indicates that the square root of the entire AVE for latent variables is greater than the correlation coefficients with other determining factors, thereby confirming the good discriminant validity of the analysis (Fornell & Larcker, 1981). Besides considering the Fornell–Larcker test (Fornell & Larcker, 1981; Hair et al., 2021), also proposed to observe the value of HTMT (heterotrait–monotrait ratio of correlations), to highly analyze discriminant validity specifically.

In this approach, DV is considered good when the HTMT value does not exceed the 0.9 threshold. Using smart-PLS, Table 6 also reveals that the highest HTMT value is 0.898 (PE-BI), proving that the DV between the latent variables is good.

4.3. Structural model

Before the structural model analysis, the values of VIF (Variance Inflation Factor) need to be evaluated to analyze collinearity between indicators. To prove the unbiased nature of the model, the VIF value should not be more than 10 (Hair et al., 2009). According to Table 3, the maximum and average VIF values are 9.042 and 5.626, respectively. This confirms the absence of significant collinearity issues among the constructs, validating that the initial model is not distorted or inaccurate.

Based on these results, it can be inferred that the model used does not suffer from bias problems. Additionally, values such as RMS_theta, NFI, and Standardized Root Mean Residual (SRMR) are commonly used as PLS-SEM indicators to assess the overall adequacy of the model. SRMR values are observed between 0 and 1, and when <1.00, the SRMR value is considered indicative of a well-fitting model (Wijaya et al., 2022). Therefore, the result indicates a value of 0.026 for SRMR, suggesting a good fit.

Moreover, higher NFI values above 0.9 are considered indicative of a good fit for better model performance (Hu & Bentler, 1998) (Table 7).

When evaluating the structural model, it is essential to analyze path relationships, effect sizes, R^2, and the predictive relevance of the model, as suggested by Sarstedt et al. (2022). To analyze this model and validate hypotheses, a sample survey with a sample size of 404 was conducted to examine the significance of path coefficients. The results are subsequently presented in Figure 3, illustrating the influence of PLS-SEM factors on designers' intention to use AIGC tools.

Table 8 indicates that H8, H7, H12, H1, H10, H14, H4, H5, H6, and H13 have reached significance levels, with *p*-values <0.001. On the other hand, H2, H3, H9, and H11 are not significant due to *p*-values exceeding 0.05. The model's latent variables show no strong collinearity (VIF <5), leading to the following conclusions: (1) the structural items do not overlap, (2) each item independently reflects the measured indicators, and (3) the questionnaire design is reasonable.

Furthermore, the *R*-square value is often used to describe the explainability of a model, with the value commonly observed between 0 and 1 (Shiferaw et al., 2021). This indicates that a higher *R*-square value led to greater explanatory power. Based on the results, 76.2, 39.3, 4.8, 67.9 and 23.8% of the BI, EE, FC, PE and PR variances toward intention to use AI painting generation (Table 9).

5. Discussion

In this study, we conducted a questionnaire survey among design professionals and students to explore the behavioral intentions of designers towards AI-generated content (AIGC) tools and the influencing factors. We formulated a total of 14 hypotheses and drew conclusions through data

Table 5. Fornell–Larcker test for discriminant validity test.

Fornell–Larcker test for discriminant validity test.

	BI	EE	FC	PA	PE	PR	SI
BI	0.941						
EE	0.698	0.953					
FC	0.236	0.384	0.929				
PA	−0.245	−0.350	−0.189	0.969			
PE	0.838	0.792	0.297	−0.284	0.935		
PR	−0.632	−0.562	−0.258	0.334	−0.598	0.974	
SI	0.687	0.539	0.224	−0.243	0.623	−0.489	0.960

Table 6. HTMT (heterotrait–monotrait ratio of correlations) values.

HTMT (heterotrait–monotrait ratio of correlations) values.

	BI	EE	FC	PA	PE	PR	SI
BI							
EE	0.740						
FC	0.253	0.409					
PA	0.257	0.365	0.201				
PE	0.898	0.843	0.320	0.300			
PR	0.663	0.584	0.271	0.344	0.629		
SI	0.726	0.565	0.237	0.253	0.661	0.507	

Table 7. Structural Equation Modeling (SEM) statistics.

	Saturated model	Estimated model
SRMR	0.026	0.099
d_ULS	0.161	2.256
d_G	0.318	0.354
Chi-square	804.895	774.470
NFI	0.923	0.926

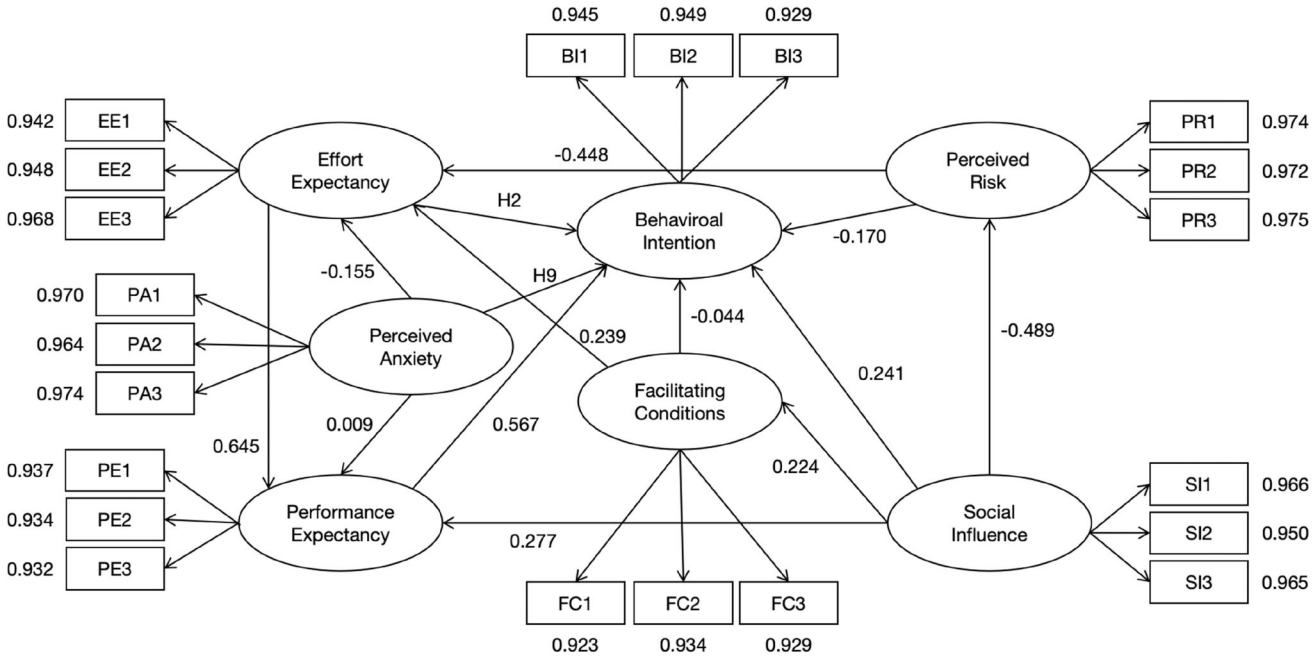


Figure 3. Path analysis results.

Table 8. Summary of hypothesis testing results.

	VIF	Path coefficient	Mean (M)	STDEV	*T* statistics	*p* Values
EE->BI	3.030	0.056	0.055	0.060	0.922	0.357
EE->PE	1.519	0.645	0.645	0.056	11.452	0.000
FC->BI	1.182	−0.044	−0.044	0.024	1.827	0.068
FC->EE	1.085	0.239	0.239	0.039	6.135	0.000
PA->BI	1.183	0.042	0.043	0.026	1.662	0.097
PA->EE	1.140	−0.155	−0.155	0.036	4.271	0.000
PA->PE	1.145	0.009	0.008	0.030	0.285	0.776
PE->BI	3.314	0.567	0.570	0.071	8.032	0.000
PR->BI	1.713	−0.170	−0.168	0.040	4.216	0.000
PR->EE	1.178	−0.448	−0.448	0.035	12.846	0.000
SI->BI	1.705	0.241	0.240	0.048	5.002	0.000
SI->FC	1.000	0.224	0.224	0.052	4.274	0.000
SI->PE	1.417	0.277	0.277	0.054	5.124	0.000
SI->PR	1.000	−0.489	−0.489	0.045	10.807	0.000

Table 9. Model fit assessment

	R-square	*Q*-square
BI	0.762	0.670
EE	0.393	0.357
FC	0.048	0.039
PE	0.679	0.589
PR	0.238	0.225

analysis. In the following sections, we summarize and discuss the research findings.

5.1. Direct influencing factors on behavioral intention

Firstly, examining the results through the lens of standardized path coefficients, our study reveals significant impacts of performance expectancy (H1: 0.567), social influence (H4: 0.241), and perceived risk (H10: −0.170) on behavioral intention. This indicates that designers' expectations regarding the performance, social influence, and perceived risk associated with AI-generated content (AIGC) tools are pivotal determinants of their behavioral intentions.

Designers believe that the use of AIGC tools will enhance their performance, signifying a confidence that this technology can yield positive outcomes in the design process (Lawson, 2006). This positive expectation fuels designers with optimism, as they anticipate creating higher quality designs, improving work efficiency (Wang & Dong, 2023), and potentially gaining more opportunities and recognition. Such positive emotions directly prompt their willingness to embrace this technology, striving to achieve the anticipated level of performance.

When designers observe their peers or individuals within their social networks actively and beneficially using AI-generated content (AIGC) tools, they may lean towards adopting this technology due to social identity and pressure dynamics (Cheung & Lee, 2010). They likely aim to stay competitive within their professional circles, keep abreast of technological advancements, and reap similar benefits. The emotional impetus of social identification directly influences their willingness to use the technology.

This social influence extends beyond mere observation and includes a desire to align with prevailing practices and standards within their professional or social communities. The motivation to conform, gain recognition, and avoid potential ostracism may contribute significantly to designers' decisions to incorporate AIGC tools into their design workflows. Thus, social dynamics play a crucial role in shaping designers' perceptions and intentions related to technology adoption.

5.2. Assuming the underlying mechanisms of support and non support

While the effort expectancy (H2) did not directly yield a significant impact on behavioral intention, the positive influence of effort expectancy on performance expectancy (H8) found support. This discovery unveils a profound logical

chain: designers anticipating better performance by investing more effort in using AI-generated content (AIGC) tools tend to believe in the technology's latent value. (Goldschmidt, 1995). The expectation of effort may inspire a belief among designers that additional input can enhance their creativity, efficiency, or innovation in design (Roskes et al., 2012), ultimately leading to better performance. The anticipation of these positive effects may drive designers to be more willing to explore and adopt this technology.

Simultaneously, the positive impact of effort expectancy on performance expectancy might influence designers' technological attitudes. Designers may consider the expected performance of technology use as a crucial factor in decision-making, believing that the effort invested can directly translate into superior outcomes. This perception may further shape designers' behavioral intentions. The anticipation associated with effort expectancy may involve enhancing one's knowledge and skills to adapt and master new technology (Dahlman & Westphal, 1981) Designers may believe that by exerting additional effort, they can better understand and apply AIGC technology in practical design, leading to improved performance. This belief in skill enhancement may play a positive role in the decision-making process for designers.

Understanding how effort expectancy influences behavioral intention through its impact on performance expectancy is crucial for both designers and technology developers. Designers can gain a clearer understanding that investing extra effort can result in improved performance, encouraging them to actively explore and apply new technologies. Technology developers can emphasize the actual effects and value of the technology to increase designers' confidence in its adoption.

5.3. Complexity of social network influence

The positive impact of social influence on behavioral intentions underscores the significant role of social networks in designers' acceptance of new technology. However, the positive effects on convenience conditions and performance expectancy may be more intricate. This complexity may arise because social networks not only influence individual attitudes (Verma & Sinha, 2018; Strohmaier et al., 2019) but may also shape their perceptions of technological characteristics through information sharing and knowledge dissemination. Peers within a community might share information such as usage techniques and problem-solving methods (Bobrow & Whalen, 2002), making it easier for designers to grasp the usage methods of the technology and thereby enhancing their perception of the tool's convenience.

Research also indicates that social influence has a negative impact on perceived risk. In many ways, information sharing can help users and communities reduce anxiety due to a lack of understanding of technology information, as it can reduce uncertainty (Hamilton et al., 1994). Peers within a community may share their positive views and successful experiences with the technology, thereby alleviating designers' concerns about potential negative impacts of the technology. This positive information sharing within the community may reduce the perceived level of risk. However, peers within a community may also share views on potential issues, risks, and ethical considerations of the technology, influencing designers' perception of technological risks. Negative information sharing within the community may deepen the perceived risk, or conversely, positive information sharing in a supportive community may alleviate perceived risks (Hussain et al., 2018).

Considering the complexity of social network influence is crucial for the practical application of technology acceptance and adoption. Technology developers can actively utilize success stories within communities to propagate positive influences, thereby strengthening designers' confidence and attitudes toward the technology. Simultaneously, designers can engage in community interactions to acquire valuable information and advice, addressing potential technological challenges and difficulties.

5.4. The multifaceted role of perceived risk

Perceived risk plays a multifaceted role in designers' willingness to use AIGC-assisted design tools, involving cognitive, emotional, and behavioral factors in the process of technology acceptance.

Perceived risk plays a crucial role in the cognitive process of designers. When faced with the use of AIGC technology, designers may have concerns about ethical issues, privacy concerns, copyright problems, and more (Tao et al., 2023; Wu et al., 2023; Zhang et al., 2023). This constitutes cognitive risk. These risks may form a negative framework for technology use in the minds of designers, influencing their overall perception of the technology and even leading to considerations about the existence of issues related to AIGC (Wu et al., 2023). This cognitive risk may be correlated with the path coefficient H13, indicating that social influence influences perceived risk.

Perceived risk also involves impacting users' emotional experiences (Lang, 2018). Designers may feel anxious and concerned about potential negative consequences of the technology, constituting emotional risk. Emotional risk affects individuals' attitudes and emotions (Strohmaier et al., 2019), lowers behavioral intentions (Bae & Chang, 2021), making them more resistant to adopting the technology. This emotional risk may be correlated with the path coefficient H10, indicating that perceived anxiety negatively influences performance expectations.

Perceived risk plays a decisive role in the technological decision-making process of designers. Designers may weigh different risk factors, evaluating the potential benefits and risks of the technology (Wu et al., 2023). Perceived risk may lead designers to approach technology more cautiously or even abandon its use. The impact and moderation effect of social influence may be correlated with the path coefficient H14.

A deeper understanding of the multifaceted role of perceived risk in the process of technology acceptance provides important insights for technology promotion and adoption (Lee, 2009). Technology developers need to be aware of designers' concerns about risks and mitigate these concerns through transparent information communication and risk

management measures. Simultaneously, designers need to consider the long-term value and potential risks of technology in the risk balancing and decision-making process.

5.5. The indirect role of perceived anxiety

Perceived anxiety plays an indirect yet crucial role in the formation of designers' willingness to use AIGC tools. Although perceived anxiety itself does not directly impact behavioral intentions, its significant indirect effects on performance expectations and effort expectations result in substantial consequences for designers' actual decisions.

Firstly, concerning performance expectations, designers' willingness to use AIGC tools is positively influenced by their expectations of performance improvement. However, the presence of perceived anxiety has a negative moderating effect on this positive relationship (Heron & Smyth, 2010). This indicates that even if designers anticipate that AIGC tools can enhance their work efficiency and creative output, perceived anxiety may still act as a constraining factor, reducing their actual adoption intentions towards this technology.

Secondly, regarding effort expectations, designers' expectations of effort required for using AIGC tools are also significantly influenced by perceived anxiety. The anticipated effort required for learning and adapting to new technology may decrease due to perceived anxiety (Beaudry & Pinsonneault, 2005), affecting their willingness to adopt AIGC tools. This finding further underscores the complex indirect impact mechanism of perceived anxiety in shaping designers' attitudes towards AIGC tools.

These two aspects of results suggest that when promoting AIGC tools, it is crucial not only to enhance designers' expectations of tool performance but also to address and alleviate perceived anxiety. This approach can encourage designers to embrace and apply new technology more actively. In future research and practice, exploring effective strategies to reduce perceived anxiety can provide comprehensive support for the promotion of AIGC tools in the design field.

5.6. Explanation for non-significant relationships

Some hypotheses were not supported, such as the impact of effort expectations on behavioral intentions (H2) and the influence of convenience conditions on behavioral intentions (H3). This may be related to various factors such as sample characteristics and research background. In future research, exploring the potential mechanisms behind these non-significant relationships can provide a more comprehensive explanation.

As for the hypotheses that were not supported regarding the impact of perceived anxiety on behavioral intentions (H9) and the influence of perceived anxiety on performance expectations (H12), this study analysis suggests that firstly, the influence of perceived anxiety may be related to the designers' level of experience. Experienced designers may have a more mature judgment of the potential risks of new technology, thereby mitigating the degree of perceived anxiety. Conversely, relatively novice designers may be more susceptible to negative information, resulting in stronger perceived anxiety (Hinds, 1999). Secondly, risk sensitivity may affect individuals' reactions to perceived anxiety (Bardeen et al., 2013). Some designers may be inherently more cautious and sensitive to potential risks, making them more prone to experiencing perceived anxiety. On the other hand, some designers may be more optimistic, inclined to see the positive aspects of technological outcomes, thereby reducing the level of perceived anxiety. Thirdly, emotional attitudes and mood states may influence individuals' responses to survey questions (Podsakoff et al., 2003) Designers in a positive emotional state may be more willing to accept new technology, alleviating the negative emotions generated by perceived anxiety. Conversely, designers with poor emotional states may be more susceptible to the impact of perceived anxiety.

5.7. Significant moderating effect of gender

Encouragingly, our study reveals a significant moderating effect of gender on the relationship from effort expectations to performance expectations. Specifically, for female designers using AIGC-assisted design tools, the coefficient from effort expectations (EE) to performance expectations (PE) is 0.234 higher compared to male designers. This suggests that female designers are more inclined to translate the anticipated benefits of technology into actual adoption intentions. In comparison to males, their expectations of AIGC tools significantly influence the actual adoption behavior. This gender difference may be influenced by various factors, such as individuals' attitudes toward technology, confidence, and expectations for future use, necessitating further exploration in future research.

Gender moderation effect			
	Difference (female − male)	1-Tailed (female vs. male) *p* value	2-Tailed (female vs. male) *p* value
EE ->BI	−0.053	0.657	0.686
EE ->PE	0.234	0.013	0.025
FC ->BI	0.030	0.267	0.534
FC ->EE	−0.153	0.973	0.054
PA ->BI	0.031	0.278	0.556
PA ->EE	−0.009	0.548	0.905
PA ->PE	−0.080	0.898	0.204
PE ->BI	0.008	0.479	0.958
PR ->BI	−0.034	0.657	0.686
PR ->EE	−0.118	0.950	0.099
SI ->BI	−0.014	0.558	0.884
SI ->FC	−0.003	0.512	0.977
SI ->PE	−0.176	0.960	0.080
SI ->PR	0.006	0.473	0.946

5.8. Experience's significant moderation effect

Another crucial moderating variable is the designers' level of experience. We observed a significant moderation effect of experience on the relationship between perceived risk (PA) and behavioral intention (BI). Specifically, experienced designers exhibit a coefficient on the path from perceived risk to behavioral intention that is 0.173 higher than that of inexperienced designers. This suggests that the assessment of perceived risk

more strongly influences actual behavioral intentions among experienced designers. The moderating effect of experience might stem from the fact that experienced designers possess a better understanding and evaluation capability of the potential risks associated with technology, instilling them with greater confidence in addressing these concerns. Future research could delve deeper into how experienced professionals in the design field adapt and adopt new technologies more flexibly.

Experience in adjusting effects			
	Difference (we – woe)	1-Tailed (we vs. woe) *p* value	2-Tailed (we vs. woe) *p* value
EE -> BI	0.206	0.147	0.294
EE -> PE	0.060	0.400	0.801
FC -> BI	0.017	0.368	0.737
FC -> EE	0.134	0.056	0.113
PA -> BI	0.173	0.007	0.015
PA -> EE	0.106	0.280	0.560
PA -> PE	0.074	0.310	0.619
PE -> BI	−0.155	0.838	0.324
PR -> BI	−0.030	0.616	0.769
PR -> EE	0.099	0.272	0.543
SI -> BI	−0.062	0.680	0.640
SI -> FC	0.143	0.193	0.386
SI -> PE	0.012	0.496	0.991
SI -> PR	0.003	0.492	0.984

5.9. Non-significant moderating effects of other variables

Although we examined age in our study, no significant moderating effects were found in any of the relationships. This may be because the influence of age on designers' adoption behavior of AIGC-assisted design tools is relatively weak, or the age differences in the sample are not sufficient to be statistically significant. In future research, it might be worthwhile to consider expanding the age range or exploring different dimensions of age factors to more comprehensively understand the potential impact of age on technology adoption behavior.

Moderating effects of age			
	Difference (old – young)	1-Tailed (old vs. young) *p* value	2-Tailed (old vs. young) *p* value
EE -> BI	−0.108	0.763	0.473
EE -> PE	0.024	0.396	0.792
FC -> BI	0.124	0.057	0.114
FC -> EE	−0.224	0.972	0.055
PA -> BI	0.150	0.088	0.176
PA -> EE	−0.148	0.869	0.263
PA -> PE	−0.029	0.622	0.757
PE -> BI	0.184	0.137	0.273
PR -> BI	−0.073	0.671	0.659
PR -> EE	0.020	0.415	0.831
SI -> BI	−0.128	0.899	0.203
SI -> FC	−0.076	0.686	0.628
SI -> PE	0.074	0.318	0.635
SI -> PR	0.168	0.125	0.250

6. In conclusion and for future research

When considering the combined influence of performance expectations, social influence, and perceived risks on designers' willingness to use AIGC-assisted design tools, we draw the following conclusions: Designers' willingness to use AIGC-assisted design tools is influenced by a complex interplay of various factors. Performance expectations stimulate positive anticipations among designers, fostering the belief that the technology can enhance their work performance. Social influence, driven by social identification and competitive pressure, steers designers toward a greater inclination to adopt this technology. However, perceived risks elicit negative emotions among designers, as concerns about potential ethical and legal issues may impose constraints on their willingness to use the technology. These factors interact across cognitive, emotional, and behavioral dimensions, collectively shaping the ultimate decisions of designers.

While this study aimed to provide comprehensive insights, it has some limitations. Firstly, sample selection may introduce a degree of selection bias. Secondly, the questionnaire survey may be subject to recall bias and subjective evaluations.

Future research could consider introducing voluntariness as a moderating variable to comprehensively understand the motivation behind designers' adoption of AI-generated content (AIGC) for assisted design tools. Voluntariness may play a crucial role in the process of technology adoption, influencing individuals' willingness and attitudes.

In practical applications, understanding these influencing factors can guide technology developers and organizations in formulating more effective strategies for promotion and training to increase designers' acceptance and use of AIGC tools. Given the ongoing developments in the design field, future research could expand to other design tools and domains, providing broader insights and understanding. In summary, this study reveals the intricate cognitive, emotional, and behavioral factors behind designers' willingness to use AIGC tools, offering valuable insights for advancing technological innovation and design practices. Subsequent research will further deepen our understanding of these influencing mechanisms, facilitating the better application and promotion of technology in the design domain.

Disclosure statement

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